

INVITATION DETAILED: DEEP BIO ACCELERATOR FOUNDER INTERVIEW

Invitation received from the Accelerator Success Team at HomeLab (Powered By LabFellows) for the Founder Interview. Focus indicates Launch Track parameter alignment (\$0M - \$2M funding targets).

Host: HomeLab Team (startup@homelab.com) | Format: 30-Min Video Call (All Co-founders Required)

1. Executive Summary & Value Proposition

Nilus Lab is commercializing high-impact science to reverse cardiac ageing without tumor risk (oncogene-free). Our proprietary platform, Zenith v28.0, leverages a deep foundation manifold trained on 500,000+ single cardiac cells from the Human Cell Atlas (HCA) and the PERIHEART dataset. By utilizing variational autoencoders (scVI) and in-silico modeling & simulation tools, Zenith identifies combinatorial gene networks that restore youthful chromatin profiles and contractility in senescent cells, moving biology out of the lab and into the clinic.

2. Technology & Platform Validation (Traction)

In-Silico Modeling: Trained on 500k integrated cells across 4,908 Highly Variable Genes (HVGs) using scVI deep learning to compress 50GB matrices into 53.7MB neural weights.

Epigenetic Reset: Achieved a deterministic -11.9 year epigenetic age reversal in mature cells verified via molecular clocks.

Safety Gating: Built-in safety ceilings automatically filter out oncogenes and restrict dedifferentiation to protect functional cell identity.

AF3 Valency Audit: Integrates automated AlphaFold 3 structured motifs to verify target binding affinity (e.g., POU5F1-SOX2 complex).

3. Competitive Landscape

Retro Bio & Altos Labs: Focus on broad, systemic rejuvenation which requires hundreds of millions in funding. Nilus targets tissue-specific (cardiac) indications with lean, computational efficiency.

Turn Biotechnologies: Utilizes mRNA-based epigenetic restoration but focuses on dermatology and skin indications. Nilus is optimized specifically for the cardiac lineage.

The Nilus Edge: In-silico dosage networks predict transient transcription factor windows (e.g., 2h ON / 21h OFF) to resolve oncogene and cell drift risks before lab synthesis.

4. Founder Interview Preparation Guide (Julio de Unamuno IV Focus)

The Founder Interview is a highly focused 30-minute video call with Julio de Unamuno IV and the HomeLab team. Julio's background is centered on bringing high-impact science out of the lab and into the market. Key preparation points:

In-Silico to Wetlab Transition: Julio values in-silico modeling and simulations (Discovery Studio, Maestro). Be prepared to explain how Zenith's predictions directly guide your wetlab assay setups.

Market indications & TAM: Position Nilus against the \$50B longevity market. Initial clinical focus is on cardiomyopathy and age-induced heart failure, productized as an out-licensing platform for pharma.

Competition & Barriers: Demonstrate how Nilus bypasses Altman-style reprogramming risks (oncogenesis, cell drift) computationally, creating strong IP and high safety barriers compared to Altos/Retro.

Team Chemistry & Dynamics: Since all co-founders are required to join, show the interdisciplinary synergy across computation (VAE architectures), systems engineering, and molecular biology.

Accelerator Goals: Secure next round of funding (\$0M - \$2M Launch Track parameters), leverage HomeLab lab networks for early animal trials, and target a cohort presentation at the BIO 2026 Innovation Summit in San Diego.

5. HomeLab Contact & Follow-up Details

The video call is scheduled using the booking link in the invite. In case of schedule shifts or coordinator follow-ups, the Accelerator Success Team coordinates via HomeLab Powered By LabFellows:

- Email contact: startup@homelab.com
- Tel contact: +1 (833) 452-2677 ext. 500
- HomeLab homebase website links local tools and preparation documents.